Indian Institute of Technology, Bhilai

CSP203: Software Tools & Technologies Laboratory

Lab Activity - 18

Full Marks: 45 *Deadline: 11:59 PM, 18-11-2025*

Submission Instructions:

1) Answer all the questions.
2) You should prepare a report which contains screenshots of code and output of the task. The report contains **5 marks.**
3) Your report file should be named as <Roll No>_<Name>_<Lab Activity No>.pdf
4) Example: B24CS055_Shareen_18.pdf
5) **Plagiarism is strictly prohibited.**

Objectives:

1) To understand the fundamentals of Docker image and container management.
2) To gain hands-on experience in interacting with running containers.
3) To develop practical skills in building custom Docker images using Dockerfiles.
4) To learn volume usage, networking, and system-level inspection tools.
5) To strengthen the understanding of containerized workflows and improve readiness for deploying applications using Docker in real-world environments.

Question - 1: [10 Marks]

Image Management and Transfer

Perform the following Docker image operations:

1. Pull any official image from Docker Hub using docker pull.
2. List all local images using docker images.
3. Display the history of the pulled image using docker history.
4. Inspect the image metadata using docker image inspect.
5. Save the image to a tar archive using docker save.
6. Remove the image using docker rmi.
7. Load the image back into Docker using docker load

Submit commands and screenshots for each step.

Question - 2: [10 Marks]

Container Creation and Interaction

Using the ubuntu image:

1. Create and run a container in **interactive mode** using docker run -it.
2. Install any package inside the container (e.g., curl, vim, or git).
3. Open another terminal and check running containers using docker ps.
4. Use docker exec to run a command (ls, hostname, etc.) inside the running container.
5. Stop the container with docker stop and remove it using docker rm.

Provide commands and screenshots.

Question - 3: [10 Marks]

Building and Running a Custom Docker Image (Dockerfile)

Create a Dockerfile that includes:

1. A base image (Ubuntu, Alpine, or Python).

2. An environment variable using `ENV`.
3. A working directory using `WORKDIR`.
4. Copying source files using `COPY`.
5. Installing software using `RUN`.
6. Exposing a port using `EXPOSE`.
7. A default command using `CMD` or `ENTRYPOINT`.

Then:

1. Build the image using docker build.
2. Run a container using this custom image.
3. Inspect the image using docker image inspect.
4. Display image history using docker history.

Submit the Dockerfile, build logs, and screenshots.

Question - 4: [10 Marks]

Docker Container Logs, Inspection, and File Operations

Perform the following tasks using any container (e.g., Nginx, Ubuntu, or Alpine):

1. Run a container in detached mode using docker run -d.
2. Display container logs using docker logs.
3. Copy a file **from the host to the container** using docker cp.
4. Copy a file **from the container to the host** using docker cp.
5. Show running processes inside the container using docker top.
6. Inspect the container metadata using docker inspect.
7. Finally stop and remove the container.

Provide all commands used and screenshots of the results.